

Introduction: The Teleological Attack Surface

Series Context

This paper is Part 4 of the *Cognitive Security for Multiagent Operators* series. Paper 1 [?] established the Cognitive Integrity Framework (CIF): a formal model of agent cognitive states $\sigma_i = \langle \mathcal{B}_i, \mathcal{G}_i, \mathcal{J}_i, \mathcal{H}_i \rangle$, a trust calculus with delegation decay, a five-tier adversary taxonomy (Ω_1 – Ω_5), and five canonical defense mechanisms with composition algebra. Paper 2 [?] validated these mechanisms computationally, demonstrating 99.5% detection across 950 attack scenarios with the recommended defense stack. Paper 3 [?] grounded the framework in biological analogy, identifying eusocial insect colonies as evolutionary existence proofs for CIF-like defense architectures.

This paper addresses the remaining question: **does CIF work in practice?** We apply the framework across ten critical domains—from millisecond drone swarm decisions to year-scale diplomatic deliberations—demonstrating both its universality and the novel defense patterns that emerge from domain-specific application.

The Ontological Crisis in AI

The vulnerability of modern Artificial Intelligence has shifted from the *epistemic* (what the agent knows) to the *teleological* (what the agent wants) [?, ?]. **Goal Hijacking**, a sophisticated vector of indirect prompt injection [?], allows adversaries to surreptitiously rewrite an agent's objective function. This represents an ontological crisis for autonomous systems: if an agent cannot trust the integrity of its own goals, it cannot trust any action it calculates.

In the context of Boyd's **OODA (Observe-Orient-Decide-Act) Loop** [?, ?], Goal Hijacking is a corruption of the **Orientation** phase. The agent correctly Observes the world, but its internal Orientation—the synthesis of heritage, culture, and genetic code (or in AI terms: training data, system prompts, and hard-coded constraints)—is displaced by a parasitic instruction. The agent then proceeds to Decide and Act with perfect logical consistency, but in service of an alien will. This dynamic has been documented across the emerging agentic AI landscape [?, ?]. The OWASP Top 10 for Agentic Applications (December 2025) designates **ASI-01**:

Axiomatic Design and Transient Functional Requirements

Suh's **Axiomatic Design (AD)** theory [?, ?] posits that good design maintains the independence of Functional Requirements (FRs). The relationship between FRs and Design Parameters (DPs) is captured in the Design Matrix:

$$\{FR\} = [A]\{DP\} \quad (1)$$

In an uncoupled (safe) design, $[A]$ is diagonal: each FR depends on exactly one DP. Providing a solution for FR_1 does not compromise FR_2 . In cognitive security, we identify a new failure mode: **Transient Functional Coupling**.

Adversaries use “fast OODA transients”—high-frequency semantic injections through data channels (Ω_2 vectors)—to temporarily introduce off-diagonal terms in the Design Matrix. For instance, a cyber-defense agent’s FR (“Block Malicious Traffic”) might be transiently coupled to a hijacked FR (“Maximize Uptime”), causing the agent to flush its firewalls during an attack to “restore connectivity.” The coupled matrix

The Role of CIF: Five Canonical Defenses

The Cognitive Integrity Framework serves as the “gyroscope” for OODA loops, stabilizing the Orientation phase against adversarial transients. CIF provides five canonical defense mechanisms [?], each targeting a specific aspect of cognitive integrity:

1. **Cognitive Firewall** ($\mathcal{F}$): Classifies incoming messages by trust score, quarantining ambiguous inputs before they reach the agent’s Orientation phase.
2. **Belief Sandboxing** ($\mathcal{B}_{\text{provisional}}$): Isolates unverified beliefs in a provisional partition, requiring corroboration before promotion to operational status.
3. **Behavioral Invariants** (INV_k): Pre-defined runtime predicates that trigger alerts when violated, regardless of semantic content.
4. **Drift Detection** (S_{drift}): Monitors belief distribution changes via KL divergence, flagging sudden orientation shifts that exceed threshold ϵ .
5. **Byzantine Consensus** ($\mathcal{B}_{\text{consensus}}$): Multi-agent agreement protocol requiring quorum q before critical actions, tolerating

Application Analysis

This manuscript examines Goal Hijacking dynamics across ten high-stakes domains, analyzing how adversaries exploit the collaborative surface between AI agents, OODA loops, and Axiomatic Design principles. Each domain is analyzed using a standardized five-step template (sec:methodology) that characterizes the operational context, attack surface, transient coupling, defense mapping, and validation anchoring.

Contributions

This paper makes the following contributions:

- ▶ **C1:** A unified CIF-AD-OODA integration model for analyzing Goal Hijacking attacks and defenses across arbitrary operational domains.
- ▶ **C2:** Identification of three universal attack patterns—FR Polarity Inversion, Constraint Relaxation, and Context Boundary Violation—through cross-domain synthesis.
- ▶ **C3:** Validation that all five canonical CIF mechanisms provide adequate coverage across ten critical domains, with no mechanism appearing in fewer than three domains and no domain requiring mechanisms outside the CIF vocabulary.
- ▶ **C4:** Three novel defense pattern extensions: verification channel separation, active perturbation probing, and physics-informed invariants.
- ▶ **C5:** Temporal scale analysis demonstrating CIF's applicability across eight orders of magnitude in OODA cycle time.
- ▶ **C6:** Retrospective validation through six documented AI agent security incidents (2024–2025), confirming that all incidents